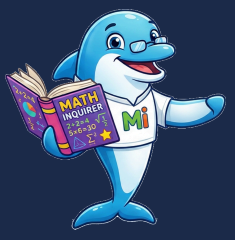
TUTOR GUIDE • LEVEL 1 • TERM 3 • LESSON 9 OF 10

Spot the Mistake



Big idea: Check with a different method; a wrong answer leaves a clue. A neat answer can still be wrong. We check it a second way, such as adding back, and we find the one step that went wrong.

90 minutes • live online or face to face • up to 8 children • ages 6-8 • Notice it. Test it. Explain why.

1. At a glance

Strand	Checking work across adding, taking away, time and money (Primary 1 process skills: reasoning and communicating) (Addition and subtraction within 100, and checking a take away by adding back (Primary 1, Addition and Subtraction, p. 31); Time: half an hour and 1 hour later (Primary 1, Time, p. 32) and money: total and left in whole dollars (Primary 1, Money, p. 31); Process skills: reasoning, communicating, and checking an answer (syllabus pp. 13 to 14)) Strand label for the tutor only. This is an added lesson (new topic): it is not a listed Primary 1 content outcome. It is error analysis across the strands of the term, and it belongs to the process skills (reasoning, communicating). Start here is Primary 1 work with scaffolding: add back to check a take away, say whether an answer is right, find the one wrong step in a worked sum of 2 or 3 steps, and spot a wrong time or total in a story (o'clock and half past only; whole dollars; numbers within 100). BEYOND PRIMARY 1 ON PURPOSE at Think harder and Go further only: two ways to check the same sum, a wrong 'fewer than' sum, a subtraction done by taking away one too many, 'Does It Make Sense?' (telling that an answer cannot be right by thinking about how big it must be, using the words about and near, never round: estimating is a Primary 4 outcome), spotting the wrong digit in a column sum, and writing a sum with one mistake for a partner. Owner decisions, 22 and 23 Oct 2026: strong content with scaffolding, said plainly to families; every lesson names a beyond item; Start here stays approachable. Do not describe the lesson as MOE-aligned. Wording rule (Term 1 audit, pre-flight D18): wrong working is always 'a child' or 'a child in another class', never a cast name. Non-routine types (tutor only): explain the wrong step, catch a wrong 'fewer than' sum, spot the wrong digit in a column sum, estimate to test an answer.
Objectives	• Check a take away by adding back, and an add by taking away, and say whether an answer is right. • Find the one wrong step in a worked sum, say what the right step is, and fix the steps after it. • Spot a wrong time (o'clock and half past) and a wrong total or amount left in a money story. • Think harder: two ways to check the same sum, a take away done by taking away one too many, a wrong 'fewer than' sum, and a time that cannot be shown on a clock. • Go further: tell that an answer cannot be right by thinking about how big it must be, spot the wrong digit in a column sum, and write a sum with one mistake for a partner.
Joining cold	**Remember warm-up covers:** Check a take away by adding back (Term 1 Lesson 7, taught again with a number line); Add tens with a jump (Term 1 Lesson 6, taught again with a number line); One hour later on a clock (Term 3 Lesson 3, taught again with a picture of a clock). Only counting to 100 in tens and ones. Nothing from the lessons before is needed: the warm-up adds back on a number line, makes a tens jump and shows a clock one hour later. The words check, wrong step and right step are taught in Explore and the Mini-lesson. Fairly safe to miss: this lesson applies Lessons 6, 7, 23 and 24, and Mi's Challenge Quest has a gate on this idea (a sum that is wrong, with the right answer). A child who misses it watches the recording and does workbook pages 1 and 2 before the Quest. A child who joins here cold can follow: the warm-up re-teaches adding back, the tens jump and one hour later, and Start here keeps to numbers within 100.

Child materials	• Your Term 3 workbook, pages 1 to 5 • The mistake-card page (page 6 of your book): a grown-up cuts out the 8 cards beforehand. The RIGHT and WRONG mat on it is not cut • No cards? Write each sum on a small piece of paper, and make two piles on the table • Your clock from the first lesson of Term 3 is a help for the time questions, but it is not needed • A pencil, a rubber and spare paper
Tutor materials	• The slide deck • Your own set of mistake cards to show on camera, if you wish • A visible clock No ruler is needed. Nothing needs to be bought. Small pieces stay on the table: nothing goes in the mouth or ears; keep them away from younger brothers and sisters. No matches, toothpicks or anything sharp. Nobody cuts during the lesson; a grown-up cut the cards beforehand. The cards show sums with one mistake on purpose: wrong working is never given a name, and no child is told they made it.
Tiers	Start here (6 questions, pages 1 and 2), Think harder (4, page 3), Go further (3, page 4), Investigation and look-back (page 5), kit page (page 6). Labels are neutral; nobody is told which page they are on.
Beyond Primary 1 (tutor only)	Does It Make Sense? at G1: Telling that an answer cannot be right by thinking about how big it must be (52 + 39 is about 50 + 40, so 191 is far too big; the right answer is 91), using the words about and near, never round. Estimating an answer is a Primary 4 outcome (the rounding and estimating strand), so it is beyond the school Primary 1 syllabus on purpose. The other Go further items (the wrong digit in a column sum and writing a sum with one mistake) are checking and explaining tasks. Named puzzle: Does It Make Sense? Never in Start here, never advertised to children.
Contest-style move	Used at Think harder, Go further and the Investigation: find the one wrong step and say why (explain a worked sum); check a take away by adding back, and check an add by taking away (a second method); catch a wrong 'fewer than' sum (the one with more is found by adding); spot the wrong digit in a column sum; about how big: an answer that cannot be right (estimate with about and near). Never advertised to children or parents.

Before the lesson

- Solve every workbook question yourself first. Check the answers: S1 41, Yes; S2 60, No, 27; S3 step 2, 64; S4 No, 3:30; S5 No, $13; S6 True, False, True, False; T1 step 2, 45; T2 48, 58, 85; T3 No, 34; T4 B, 8:00; G1 about 90, No, 91; G2 Tens, 85; G3 any sum with one mistake; I1 no (44), forgets the ten.
- Tell parents a week before: the mistake-card page (page 6 of the book) needs 8 cards cut out by a grown-up, and the mat is not cut. The next lesson, Calendar Detective, holds Mi's Challenge Quest: the Quest Log stays closed in its envelope, and each child needs a quest partner.
- Have the slides open. Have your own mistake cards to show on camera if you wish. Practise the arm-swing break.
- Check the recording captures the tutor and the shared screen. Children video is excluded; any unmuted voice is recorded. Greet each child by first name as they arrive, then start the recording, then say the notice; avoid first names while it runs.
- Be plain with families: checking by adding back, finding a wrong step and wrong times and totals in stories are Primary 1 work with help; telling that an answer cannot be right by thinking about how big it must be (estimating, with the words about and near, never round) and spotting the wrong digit in a column sum are beyond Primary 1 on purpose. Do not call the lesson MOE-aligned.
- Wording rule: wrong working belongs to 'a child' or 'a child in another class', never to a cast name and never to a child in the group. Say 'the sum has a mistake'. Slides keep the answers off the screen while you ask: read every question with its numbers aloud, and the three options of E2.

Safeguarding and recording (this lesson)

- The mistake cards are paper or card. Small pieces stay on the table: nothing goes in the mouth or ears. No matches or toothpicks. Nobody cuts during the lesson; a grown-up cut the cards beforehand.
- This lesson uses the main room only: no breakout rooms, so the tutor sees every camera. A parent sits nearby but does not give answers.
- Nobody holds paper or a card up to the camera: a gallery view shows everyone. Children tick their own work against the answer slide, answer together on a count of three, or show fingers or a thumb for a choice. There is no private chat.
- Wrong working belongs to 'a child in another class', never to a child in the group, and never to a name. The tutor says 'the sum has a mistake', not 'you made a mistake'. Never read out a child's name with a right or wrong answer. The recording has the tutor, the shared screen and any voice that is unmuted. Children may mute and keep their cameras off. A child's page is never shown.
- Movement break: children need a little clear space, soft shoes or socks on a safe floor, no stairs and no running. Arm swings are gentle; a child may stay seated and swing one arm.
- A child who is still stuck ticks 'I am still stuck' on the look-back page. A grown-up may photograph that page after class and send it to the business WhatsApp only; photos are kept for 90 days after the program ends. The tutor then helps in the open session next lesson. No private messages to children, and no personal phones.
- The look-back is never collected in front of the group or read out. A pass is always allowed.

2. Timing plan (90 minutes exactly)

Minutes	Part	Min	Slides	Script fills
0–2	**Opening**	2	1: Spot the Mistake	1:45
2–8	**Remember warm-up**	6	3: Warm-up: add back, Jump a ten, One hour later	4:49
8–12	**Sneak-peek puzzles**	4	1: Sneak-peek puzzles	3:10
12–23	**Explore**	11	5: Explore: right or wrong?, Four right, four wrong, Explore: fix the wrong cards, The right answers, What did we notice?	8:55
23–34	**Mini-lesson**	11	10: Add back to check, Add back: 81, not 71, Check with a different way, The two ways disagree, Which step is wrong?, Step 3 is wrong, A wrong time, Half an hour is 30 minutes, A wrong total, 13 dollars, not 15	8:52
34–38	**Brain break**	4	1: Tick-tock break	3:13
38–70	**Workbook**	32	5: Workbook page 1: Start here, Check page 1, Workbook page 2: Start here, Check page 2, Choose your next page	24:42
70–76	**Talk time**	6	2: Talk and explain, Back to the sneak-peek	4:30
76–82	**Exit ticket**	6	2: Exit ticket, Check the exit ticket	4:49
82–85	**Look-back**	3	1: Look-back	2:06
85–90	**Flex time and goodbye**	5	1: Well done, inquirers!	0:34
	Total	**90**		**67:25**

The script needs 67:25 of the 90:00 when children are quick. A slow class (speech 10% slower, children and slide changes 20% slower) needs 79:37. The last 5 minutes are a standing buffer: if you are behind, use them; do not start anything new. Clock checks: minute 12 (end of sneak-peek), minute 38 (start of workbook), minute 70 (talk time). If more than 2 minutes behind at minute 38, skip in this order: Start with: shorten Explore round 3 (what did we notice) to one question; Then: cut the money example (s-m5); Then: take the extra minutes from the flex time. Never skip the look-back or the camera and privacy moments.

3. Lesson script

Times show when each line should start, counted from the start of the lesson, if you keep to the script. SAY lines are spoken in your own words. REC lines are the short recording-voice cues for children who watch later. Children answer aloud, with fingers or hands; no private chat; no boards held up in front of the group.

0:00 – 2:00 Opening

script 1:45 of 2:00

0:00	SLIDE	Show slide **Spot the Mistake** (s-title)
0:20	REC	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your mistake cards. Pause or rewind any time.*
0:31	DO	Greet each child by first name as they arrive. Then start the recording. While it runs, avoid names. Check that you can see each child. [0:22]
0:53	SAY	Hello, inquirers! Please stay muted unless I invite you to speak. Cameras on if you can, so I can see you; cameras off is fine too. Pencil ready, and your mistake cards and mat on the table.
1:12	SAY	This lesson is recorded so a friend who is away can catch up. The recording has me, the shared screen and any voice that is unmuted. You may mute and keep your camera off.
1:28	SAY	Today we are mistake detectives. A neat answer can still be wrong. We check an answer a different way, and we find the one step that went wrong. Finding a mistake is clever.
	NOTE	Say the recording notice in the SAY line above in your own words after the recording has started: the recording has the tutor, the shared screen and any unmuted voice; children may mute and keep the camera off. The routine and the four inquirer rules were introduced in the first lesson of Term 1: do not re-teach them.
	NOTE	Plain statement for families when it comes up: checking by adding back, finding a wrong step and wrong times and totals in stories are Primary 1 work with help; telling that an answer cannot be right by thinking about how big it must be, and the wrong digit in a column sum, go beyond Primary 1 on purpose.
	NOTE	New here? Say in one minute: our three rule words (notice it, test it, explain why); a guess is fine and you can always pass; after Start here you may choose Think harder, Go further or the Investigation; nobody holds paper to the camera. A child without cards writes the 8 sums on paper. Wrong working in this lesson belongs to a child in another class, never to a child in the group.

2:00 – 8:00 Remember warm-up

script 4:49 of 6:00

2:00	SLIDE	Show slide **Warm-up: add back** (s-rem1)
2:15	REC	*Recording: join in! Look at each picture and say your answer out loud. Pause if you need longer.*
2:24	SAY	Take away, then check by adding back. Forty take away fifteen is twenty-five. To check, we add the fifteen back to the answer.
2:36	ASK	What is twenty-five plus fifteen? Do we get back to forty? [wait about 30 s for answers]
3:11	CHILDREN	Your turn. Fifty take away twenty is thirty. Add back to check. Say it with me. [0:25]
3:36	SLIDE	Show slide **Jump a ten** (s-rem2)
3:51	SAY	A tens jump. Forty-six plus thirty. We jump three tens, and the ones stay the same.
3:59	ASK	Where do we land after three jumps of ten? [wait about 30 s for answers]
4:34	CHILDREN	Your turn. Fifty-three plus twenty. Where do you land? [0:25]
4:58	SLIDE	Show slide **One hour later** (s-rem3)
5:14	SAY	A clock. It is three thirty. One hour later, the short hand moves to the next number. The long hand stays on the six.
5:26	ASK	What time is it one hour later? [wait about 30 s for answers]

5:59	**CHILDREN**	Your turn. It is eight o'clock. What time is it one hour later? [0:25]
6:24	**CHILDREN**	Say each answer with me: forty, seventy-six, four thirty. Then stretch your fingers. [0:25]
	IF STUCK	If a child is unsure how to add back, count on from 25 to 40 in fives on fingers. For the tens jump, three fingers for three tens. For the clock, count one hour from 3 to 4 and show the long hand stays still.
	NOTE	Answers: 40; 76; 4:30. Children answer with a whisper, fingers or hands. Nobody writes. Joining cold: all you need is counting to 100 in tens and the pictures.

8:00 – 12:00 Sneak-peek puzzles

script 3:10 of 4:00

8:00	**SLIDE**	Show slide **Sneak-peek puzzles** (s-sneak)
8:12	**REC**	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*
8:22	**SAY**	Two sneak-peek puzzles: ideas we have not learnt yet. A guess is a great start, and I do not know is welcome. Nobody is marked.
8:34	**READ**	Puzzle one. A child in another class writes forty plus twenty equals seventy. Is the answer right? Circle Yes or No.
8:45	**CHILDREN**	Children write on their own paper. [0:40]
9:25	**READ**	Puzzle two. A child in another class writes twenty-seven plus sixteen equals thirty-three. Is the answer right? I know because blank.
9:37	**CHILDREN**	Children write or draw their answer. [1:25]
11:02	**SAY**	Keep your paper. Nobody shows it. We will check our own thinking later, at Talk time.
	NOTE	Puzzle 1 answer: No (40 + 20 = 60). Puzzle 2 answer: No (27 + 16 = 43; the ten from 7 + 6 = 13 was forgotten). Look for good ideas to praise later (a child who checked by taking away, or who said 7 and 6 make 13). Do not write scores or keep a list.

12:00 – 23:00 Explore

script 8:55 of 11:00

12:00	**SLIDE**	Show slide **Explore: right or wrong?** (s-explore1)
12:15	**REC**	*Recording: pause now. Sort your 8 cards into RIGHT and WRONG. Then fix each wrong card. Press play when you have tried.*
12:26	**SAY**	The cards are the teachers today. Some sums on your cards are right and some are wrong. Check each card a different way. Then put it on RIGHT or on WRONG.
12:42	**CHILDREN**	Children check each card and sort it onto the mat. Tutor scans the cameras. [2:30]
15:12	**ASK**	Which card did you put on WRONG? How did you check it? [wait about 45 s for answers]
16:02	**SLIDE**	Show slide **Four right, four wrong** (s-explore1-ans)
16:10	**SAY**	Four are right and four are wrong. A neat answer can still be wrong, so we check.
16:19	**SLIDE**	Show slide **Explore: fix the wrong cards** (s-explore2)
16:34	**SAY**	Round two. Take the four wrong cards. What went wrong in each one? Write the right answer on your paper.
16:44	**CHILDREN**	Children fix the four wrong cards and write the right answers. [1:40]
18:24	**ASK**	Which card did you fix first? What went wrong? [wait about 40 s for answers]
19:08	**SLIDE**	Show slide **The right answers** (s-explore2-ans)
19:16	**SAY**	Seventy-three: the ten was lost. Seventy-eight: the ones were swapped round. Thirty-three: a plus sign was used. Twenty-two: one short.
19:26	**SLIDE**	Show slide **What did we notice?** (s-explore3)
19:34	**ASK**	How far was each wrong answer from the right one? Does the size of the mistake give a clue? [wait about 55 s for answers]

20:39	SAY	A wrong answer leaves a clue. A lost ten is ten away. And doing the sum the same way again can make the same mistake again, so we check a different way.
	IF STUCK	A child who cannot decide on a card: check it by adding back or taking away. A child who puts a right card on WRONG: ask them to show the check. A child with no cards writes the 8 sums on paper and makes two lists.
	NOTE	Main room only; you cannot see small cards, so ask children to say the sum and to show a thumb for right or a flat hand for wrong. Answers: RIGHT 70-30=40, 18+4=22, 62-8=54, 38+24=62; WRONG 45+28=63 (73), 84-6=82 (78), 53-20=73 (33), 17+5=21 (22). Do not say the rule before the children do.

23:00 – 34:00 Mini-lesson

script 8:52 of 11:00

23:00	SLIDE	Show slide **Add back to check** (s-m1)
23:08	REC	*Recording: keep watching and copy the examples. Rewind if you miss a step.*
23:14	SAY	A child in another class writes seventy-one take away twenty-six equals fifty-five. We check by adding back.
23:23	ASK	What is fifty-five plus twenty-six? Do we get back to seventy-one? [wait about 40 s for answers]
24:08	SLIDE	Show slide **Add back: 81, not 71** (s-m1-ans)
24:16	SAY	Fifty-five and twenty-six make eighty-one, not seventy-one. So fifty-five is wrong. Forty-five and twenty-six make seventy-one, so the right answer is forty-five.
24:28	READ	Your turn. A child writes sixty take away eighteen equals fifty-two. Add back to check. Is it right? Write Yes or No, and the right answer.
24:42	CHILDREN	Children write on their own paper. Wait 10 seconds before helping. [0:38]
25:20	SAY	Fifty-two and eighteen make seventy, not sixty. So it is wrong. The right answer is forty-two.
25:28	SLIDE	Show slide **Check with a different way** (s-m2)
25:36	SAY	Another way to check. A child writes twenty-nine plus thirty-five equals fifty-four. We jump on the line: thirty, then five.
25:46	ASK	Where do we land? Does it match fifty-four? [wait about 30 s for answers]
26:20	SLIDE	Show slide **The two ways disagree** (s-m2-ans)
26:28	SAY	We land on sixty-four. The two ways do not agree, so there is a mistake. Two different ways must agree.
26:38	SLIDE	Show slide **Which step is wrong?** (s-m3)
26:46	SAY	A sum worked in three steps. A child adds fifty-six and twenty-seven.
26:52	ASK	Check each step in turn. Which step is wrong? [wait about 40 s for answers]
27:36	SLIDE	Show slide **Step 3 is wrong** (s-m3-ans)
27:44	SAY	Step three is wrong. Eighty plus three is eighty-three. We find the one wrong step and fix it.
27:53	READ	Your turn. A child works out ninety-two take away thirty-five in two steps. Step one: ninety-two take away thirty is seventy-two. Step two: seventy-two take away five is sixty-seven. Which step is wrong? Write the right answer.
28:13	CHILDREN	Children write on their own paper. Wait 10 seconds before helping. [0:50]
29:03	SAY	Step one is wrong: ninety-two take away thirty is sixty-two. Step two is fine, but it started from the wrong number. Fix step one, then do step two again: fifty-seven.
29:18	SLIDE	Show slide **A wrong time** (s-m4)
29:26	SAY	A wrong time. The bus leaves at nine o'clock. The trip takes half an hour. A child says it arrives at nine fifty.
29:38	ASK	Is that right? Where does the long hand go in half an hour? [wait about 30 s for answers]

30:14	SLIDE	Show slide **Half an hour is 30 minutes** (s-m4-ans)
30:22	SAY	Half an hour is thirty minutes, so the bus arrives at nine thirty. Nine fifty is a mistake.
30:31	SLIDE	Show slide **A wrong total** (s-m5)
30:39	SAY	A wrong total. A pen costs four dollars and a book costs nine dollars. A child says they cost fifteen dollars altogether.
30:50	ASK	How much do they cost altogether? Is fifteen right? [wait about 35 s for answers]
31:30	SLIDE	Show slide **13 dollars, not 15** (s-m5-ans)
31:38	SAY	Four and nine make thirteen. We can check the child by taking the nine away from fifteen. That leaves six, but the pen costs four, so fifteen is wrong.
	IF STUCK	If a child cannot check by adding back, put the answer and the number taken away on a number line. If a child fixes the wrong step but not the steps after it, ask: what number does the next step start from? If a child says a neat answer must be right, ask for a second way.
	NOTE	Key words to use every time: check, add back, a different way, the wrong step, the right answer. Fixed question forms: Is the answer right? Which step is wrong? What is the right answer? Wrong working belongs to a child in another class. The slides keep the answers off the screen while you ask: read every question with its numbers aloud.

34:00 – 38:00 Brain break

script 3:13 of 4:00

34:00	SLIDE	Show slide **Tick-tock break** (s-break)
34:08	REC	*Recording: stand up and join in! Tick-tock with your arms, then say the right answer with a big voice.*
34:18	SAY	Stand up and get a sip of water. Swing your arms slowly like the hands of a clock. A big swing is one hour. A half swing is half an hour.
34:33	CHILDREN	Call quick sums for the group to answer aloud: 10 + 5, 20 + 30, 50 - 20, 7 and 3, 40 - 10. Then call one wrong on purpose (6 + 6 = 13) and the children call the right answer in a big voice. Seated children swing one arm. [2:40]

38:00 – 70:00 Workbook

script 24:42 of 32:00

38:00	SLIDE	Show slide **Workbook page 1: Start here** (s-ws1)
38:08	REC	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*
38:17	READ	Page 1, Start here. Question 1: A child writes forty-one take away seventeen equals twenty-four. Part a: check. Twenty-four plus seventeen equals blank. Part b: is the answer right? Circle Yes or No. Question 2: A child writes fifty take away twenty-three equals thirty-seven. Part a: check. Thirty-seven plus twenty-three equals blank. Part b: is the answer right? Circle Yes or No. Part c: fifty take away twenty-three equals blank. Question 3: A child adds thirty-six and twenty-eight in three steps. Look at the steps. Part a: which step is wrong? Circle one. Part b: the right answer is blank.
39:12	CHILDREN	Page 1, questions 1 to 3. Tutor walks the cameras, asks how do you know. Wait 10 seconds before helping. [8:00]
47:12	SLIDE	Show slide **Check page 1** (s-ws1ans)
47:20	READ	Question 1: the check is forty-one. Yes, it is right. Question 2: the check is sixty. No. The right answer is twenty-seven. Question 3: step two. The right answer is sixty-four. Tick your own page with a tick.
47:40	SAY	After question 3 you may choose: carry on, or try Think harder. Everyone may choose, at any time.
47:49	SLIDE	Show slide **Workbook page 2: Start here** (s-ws2)
47:57	READ	Page 2. Question 4: Ravi starts to read at the time on the clock. He reads for 1 hour. A child says he stops at 3:00. Part a: is the child right? Circle Yes or No. Part b: Ravi stops at blank. Question 5: A bag costs 9 dollars and a cup costs 4 dollars. A child says they cost 12 dollars altogether. Part a: is the child right? Circle Yes or No. Part b: they cost blank dollars altogether. Question 6: tick True or False for each sum.

48:46	**CHILDREN**	Page 2, questions 4 to 6. [8:40]
57:26	**SLIDE**	Show slide **Check page 2** (s-ws2ans)
57:34	**READ**	Question 4: No. Ravi stops at 3:30. Question 5: No. They cost thirteen dollars. Question 6: True, False, True, False. Tick your own page.
57:48	**SLIDE**	Show slide **Choose your next page** (s-choice)
57:56	**SAY**	When you have finished Start here, you can choose Think harder, Go further or the Investigation. Everyone may choose. You can swap at any time. Anything not finished is for home too.
58:12	**MOVE**	Stand and reach up for 30 seconds, then sit down and choose.
58:42	**CHILDREN**	Choice time: Think harder (page 3), Go further (page 4), the Investigation (page 5), or finish and check your own work. Tutor never says who is on which page. If Start here took the whole time, say that pages 3 to 5 are for home, and keep the time. [4:00]
	IF STUCK	Q1: add back 24 and 17. Q2: add back, then take away 23 from 50. Q3: check each step in turn. Q4: move the long hand 1 hour round; the short hand moves to the next number. Q5: add 9 and 4. Q6: do each sum another way.
	IF STUCK	A child who is still stuck ticks I am still stuck on the look-back page. The grown-up can photograph that page after class and send it to the business WhatsApp (never a personal phone). You then help in the open session in the next lesson. Never a private message to a child.
	NOTE	Think harder has a wrong step in a take away, two ways to check the same sum, a wrong fewer-than sum and a time that cannot be shown on a clock. Go further has Does It Make Sense? (beyond Primary 1 on purpose), a wrong digit in a column sum and writing a sum with one mistake. Do not announce who is on which page. Start here takes a middle child about 17 minutes, so the choice time may be short. Slow class: take the extra minutes from the flex time and do not skip the answer slides.

70:00 – 76:00 Talk time

script 4:30 of 6:00

70:00	**SLIDE**	Show slide **Talk and explain** (s-talk)
70:08	**REC**	*Recording: pause. Tell a grown-up whether checking the same way again is a good check, and why.*
70:16	**READ**	A child in another class writes forty-five plus twenty-eight equals sixty-three. She works it out again the same way and gets sixty-three again. She says it must be right. Do you agree? How could she check?
70:36	**CHILDREN**	Think about it with the cards from Explore: what could she do that is a different way? [0:50]
71:26	**CHILDREN**	Talk to the grown-up next to you or to your screen for one minute. Use the sentence starter: I think... because... [0:40]
72:06	**ASK**	Who would like to share? Volunteers first. A pass is always fine. [wait about 100 s for answers]
73:52	**SLIDE**	Show slide **Back to the sneak-peek** (s-sneakans)
74:00	**SAY**	Remember the sneak-peek? Puzzle one: no, forty plus twenty is sixty. Puzzle two: no, twenty-seven plus sixteen is forty-three. The ten was forgotten.
74:12	**ASK**	Did anyone change their mind while we worked? You can just nod. [wait about 12 s for answers]
	NOTE	Model answer: no, we do not agree. Doing the sum the same way again can make the same mistake again. A different way, such as taking away 28 from 63, gives 35, not 45, so 63 is wrong; the right answer is 73. Volunteers first; never call a child by name to answer; no tallies, no rankings. Praise the reason, not the speed.

76:00 – 82:00 Exit ticket

script 4:49 of 6:00

76:00	**SLIDE**	Show slide **Exit ticket** (s-exit)
76:08	**REC**	*Recording: pause and do the exit ticket on your own. Then press play to check it.*
76:16	**SAY**	Exit ticket. Three questions, on your own, without looking back. Write your answers on page 5.

76:24	**READ**	E1. A child writes sixty-three take away twenty-eight equals forty-five. Part a: check. Forty-five plus twenty-eight equals blank. Part b: is the answer right? Circle Yes or No. E2. A child works out twenty-eight plus fifteen in steps. Which step is wrong? Option 1: twenty-eight plus two equals thirty. Option 2: fifteen take away two equals twelve. Option 3: thirty plus twelve equals forty-two. Write its number in the bracket. E3. Lily has 20 dollars. She spends 12 dollars. A child says she has 9 dollars left. Part a: is the child right? Circle Yes or No. Part b: she has blank dollars left.
77:21	**CHILDREN**	Children answer E1 to E3 on their own. [3:10]
80:31	**SLIDE**	Show slide **Check the exit ticket** (s-exitans)
80:39	**READ**	E1: seventy-three. No. E2: option 2. E3: No. Eight dollars.
80:44	**SAY**	Tick your own ticket with a tick. Well done.
	NOTE	Nothing is collected or marked live. Children tick their own ticket. If many children miss one question, spend two minutes on it at the start of the next lesson, in the open session, no names. Do not read out any child result.

82:00 – 85:00 Look-back

script 2:06 of 3:00

82:00	**SLIDE**	Show slide **Look-back** (s-look)
82:08	**REC**	*Recording: pause and fill in your look-back box. You can pass.*
82:14	**SAY**	Look-back. Tick what was hard and what helped you. Write or draw one thing you did well. If you like, tick one thing to try next time. You can pass. Nobody reads it out. If you tick I am still stuck, ask your grown-up to take a photo of this page after class and send it to our business WhatsApp. Then we will help you.
82:46	**CHILDREN**	Children fill in the look-back box on page 5. [1:20]

85:00 – 90:00 Flex time and goodbye

script 0:34 of 5:00

85:00	**SLIDE**	Show slide **Well done, inquirers!** (s-end)
85:08	**SAY**	Well done, inquirers. Next lesson is Calendar Detective, the last lesson of Term 3, and we play Mi's Challenge Quest, a game with Mi. Please have the Quest Log closed in its envelope, a quest partner, and the seven day cards cut out by a grown-up. Goodbye, and thank you to the grown-ups.
	SLIDE	Show slide **Go further: be the detective** (s-opt) *(optional)*
	SAY	Optional, only if you are on time: a child writes eight plus seven equals fourteen, and checks it with fourteen take away seven equals seven. Does the check agree with the sum?
	CHILDREN	Short think time, then take two answers. [1:30]
	SLIDE	Show slide **If we have time: pick one** (s-menu) *(optional)*
	SAY	Spare-time menu. If the whole class is ahead, pick one or two activities from the guide (about 4 minutes each).
	NOTE	This is the standing buffer. If you are behind, finish the look-back and say goodbye. Do not start anything new. Optional slide answer: the check gives 7, not 8, so the sum has a mistake; 8 + 7 = 15. If the whole class is ahead, use the Spare-time menu (guide section 3b) instead of rushing on.

3b. Spare-time menu: if the class is ahead

- Whole class ahead: use the Spare-time menu in the Flex time and the optional Go further slide. Pick one or two activities; do not start new content.
- A few children ahead: they may choose Go further or the Investigation, which are open to everyone. Never label or name a child, and never announce who is ahead.
- Never teach anything the next lesson needs. If the class is ahead, slow down: ask 'How do you know?' and let children explain, rather than rushing on.

Optional time, never added to the 90-minute sum. It comes from the 5-minute flex buffer, from slack in the parts, or from the optional Go further slide.

Activity	Minutes	You say	Recording cue	Answer
Does adding back always work?	4 (optional)	A child says: if a take away adds back to the start number, the answer is right. If it does not, there is a mistake. Is that always true? Test it with 30 take away 12 equals 18, then with 30 take away 12 equals 20.	*Recording: pause. Test adding back on 30 take away 12 equals 18, then on 30 take away 12 equals 20.*	Yes. 18 + 12 = 30, so 18 is right. 20 + 12 = 32, not 30, so 20 is wrong. Adding back to the start number is a check that always works for a take away.
Make a puzzle for your grown-up	4 (optional)	Write a sum with one mistake. Give it to your grown-up. They check it a different way and find the mistake. Then check the right answer together.	*Recording: pause. Write a sum with one mistake and ask a grown-up to check it a different way.*	No single answer. Check the right answer by adding back or taking away.
Teach it back	3 (optional)	Pretend your grown-up or a toy has never found a wrong step. Show how 45 + 28 = 63 is 10 away from 73, and what that tells us.	*Recording: pause. Show a grown-up how a wrong answer that is 10 away tells us a ten was lost.*	63 is 10 less than 73, so a ten was lost: 5 + 8 = 13 and the ten was forgotten. The size of the mistake is a clue to what went wrong.

4. Questions, answers, hints and common mistakes

Sneak-peek puzzles (minutes 8 to 12)

Puzzle 1 (access): A child in another class writes 40 + 20 = 70. Is the answer right? Circle Yes or No.

> **Answer: No. 40 + 20 = 60.**

Puzzle 2 (think): A child in another class writes 27 + 16 = 33. Is the answer right? I know because ___.

> **Answer: No. 27 + 16 = 43. The child forgot the ten: 7 + 6 = 13, which is 1 ten and 3 ones. A check: 33 take away 16 is 17, not 27.**

Unmarked. Never put answers on boards in front of the group and never read a name with an answer. Reveal the answers where your script says (usually Talk time).

Start here (workbook pages 1 and 2)

S1 **A child writes 41 − 17 = 24.**

(a) Check: 24 + 17 = ☐☐

(b) Is the answer right? Circle Yes or No.

> **Answer: 41; Yes**

Hint (say this first): Add the answer and the 17 back. Do you get 41?

Worked solution: 24 + 17 = 41. Adding back gives the number we started with, so the answer is right.

Common mistakes: Writing 24 + 17 = 31 (a counting slip).; Saying No without adding back.

Time for a middle child: about 2.5 min.

S2 **A child writes 50 − 23 = 37.**

(a) Check: 37 + 23 = ☐☐

(b) Is the answer right? Circle Yes or No.

(c) 50 − 23 = ☐☐

> **Answer: 60; No; 27**

Hint (say this first): Add back. Do you get 50?

Worked solution: 37 + 23 = 60, not 50. So 37 is wrong. 27 + 23 = 50, so the right answer is 27.

Common mistakes: Saying Yes because 37 looks about right.; Writing 73 (adding 50 and 23).

Time for a middle child: about 3.5 min.

S3 **A child adds 36 + 28 in three steps. One step is wrong.**

Step	Working
1	36 + 4 = 40
2	40 + 20 = 50
3	50 + 4 = 54

(a) Which step is wrong? Circle 1, 2 or 3.

(b) The right answer is ☐☐.

> **Answer: Step 2; 64**

Hint (say this first): Check each step on its own. Step 1: does 36 + 4 make 40?

Worked solution: Step 1: 36 + 4 = 40, right. Step 2: 40 + 20 = 60, not 50. Step 3 then starts from 60: 60 + 4 = 64. The right answer is 64. Check: 64 − 28 = 36.

Common mistakes: Circling step 3 because it is the last step.; Writing 54 (the child's answer).

Time for a middle child: about 3.5 min.

S4 **Ravi starts to read at the time on the clock. He reads for 1 hour. A child says he stops at 3:00.**

(a) Is the child right? Circle Yes or No.
(b) Ravi stops at ____ .



Answer: No; 3:30

Hint (say this first): The short hand moves to the next number. Where does the long hand stay?
Worked solution: It starts at half past 2. One hour later the short hand moves to the next number and the long hand stays on the 6. That is half past 3, or 3:30. The child forgot the half hour.
Common mistakes: Saying Yes because the hour changed.; Writing 3:00 or 2:30.
Time for a middle child: about 3 min.

S5 **A bag costs $9 and a cup costs $4. A child says they cost $12 altogether.**

(a) Is the child right? Circle Yes or No.
(b) They cost $____ altogether.



Answer: No; $13

Hint (say this first): Altogether means add. 9 and 1 make 10.
Worked solution: 9 and 4: 9 and 1 make 10, and 3 more make 13. They cost $13. Check: 12 take away 9 is 3, and the cup costs 4, so 12 is wrong.
Common mistakes: Saying Yes because 12 is close.; Taking away: 9 take away 4 is 5.
Time for a middle child: about 3 min.

True/False statements: 25 + 30 = 55; 56 − 20 = 46; 24 + 9 = 33; 80 − 5 = 85

S6 **Tick True or False for each sum.**

Answer: True; False; True; False

Hint (say this first): Do each sum a different way. Add back to check a take away.
Worked solution: 25 + 30 = 55 is true. 56 − 20 = 36, not 46, so it is false. 24 + 9 = 33 is true. 80 − 5 = 75, not 85, so it is false.
Common mistakes: Ticking True for every sum.; Taking away the tens digit by mistake: 56 − 20 = 46.
Time for a middle child: about 3 min.

Think harder (page 3): the taught idea in a new setting

T1 **A child works out 74 − 29 in two steps.**

Step	Working
1	74 − 30 = 44
2	44 − 1 = 43

(a) Which step is wrong? Circle 1 or 2.
(b) The right answer is ☐☐.

Answer: Step 2; 45

Hint (say this first): Check the answer by adding 29 back. Do you get 74?
Worked solution: Step 1: 74 − 30 = 44, right. But 29 is 1 less than 30, so we took away 1 too many. Step 2 must add 1 back: 44 + 1 = 45. Check: 45 + 29 = 74.
Common mistakes: Circling step 1 because 30 is not 29.; Writing 44 (stopping after step 1).
Time for a middle child: about 5 min.

T2 **Two children work out 58 + 27. One child writes 75. The other child writes 85.**

(a) Check: 75 − 27 = ☐☐
(b) Check: 85 − 27 = ☐☐
(c) Who is right? Circle 75 or 85.

Answer: 48; 58; 85

Hint (say this first): If 58 + 27 is right, taking 27 away must give 58.
Worked solution: 75 − 27 = 48, not 58. 85 − 27 = 58, which is where we started. So 85 is right.
Common mistakes: Choosing 75 because 5 + 7 is 12 and the ten is forgotten.; Adding instead of taking away when checking.
Time for a middle child: about 5 min.

T3 **Mei has 28 pencils. Mei has 6 fewer pencils than Aisha. A child writes 28 − 6 = 22 to find how many pencils Aisha has.**

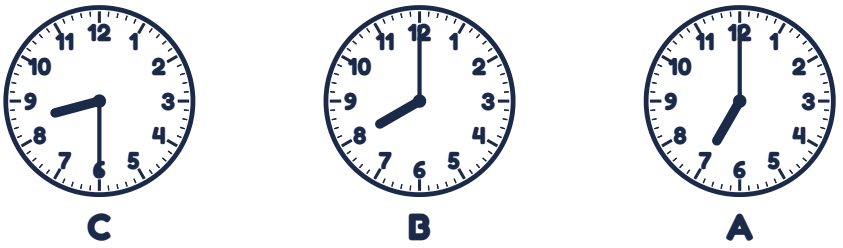
(a) Is the child right? Circle Yes or No.
(b) Aisha has ____ pencils.

Answer: No; 34

Hint (say this first): Who has more, Mei or Aisha? Draw two strips.
Worked solution: Mei has fewer, so Aisha has more. Aisha has 6 more than 28: 28 + 6 = 34. Check: 34 has 6 more than 28, and 28 has 6 fewer than 34.
Common mistakes: Saying Yes because fewer means take away.; Writing 22.
Time for a middle child: about 5 min.

T4 **Ravi leaves home at 7:30. He walks for half an hour. A child says he gets there at 7:60. Look at the clocks.**

(a) Which clock shows when Ravi gets there? Circle A, B or C.
(b) Ravi gets there at ____ .



Answer: B; 8:00

Hint (say this first): Half an hour is 30 minutes. 30 minutes after half past is on the hour.
Worked solution: A half hour after half past 7 is 8 o'clock. There is no 7:60 on a clock: after 7:59 comes 8:00. Clock B shows 8:00.
Common mistakes: Choosing clock A (7:00) or C (8:30).; Writing 7:60.
Time for a middle child: about 4 min.

Go further (page 4): mastery and extension

G1 **A child writes 52 + 39 = 191.**

(a) Do not add exactly. Circle the best guess for 52 + 39.
(b) Can 191 be right? Circle Yes or No.
(c) The right answer is ☐☐.

Answer: about 90; No; 91

Hint (say this first): 52 is near 50. 39 is near 40.
Worked solution: 52 is near 50 and 39 is near 40, and 50 and 40 make 90. So the answer is about 90. 191 is much too big, so it cannot be right. 52 + 39 = 91.
Common mistakes: Guessing about 150 because both numbers have a 9 or a 5.; Saying Yes because 191 looks like 52 and 39 joined.
Time for a middle child: about 6 min.

G2 **A child adds 47 + 38 in columns and writes 75. One digit is wrong.**

(a) Which column has the wrong digit? Circle Tens or Ones.
(b) The right answer is ☐☐.

	Tens	Ones
	4	7
+	3	8
	7	5

Answer: Tens; 85

Hint (say this first): Add the ones first. Is there a ten to carry?
Worked solution: Ones: 7 + 8 = 15. Write 5 and carry 1 ten. Tens: 4 + 3 + the 1 carried ten = 8. The child forgot the carried ten, so the tens digit should be 8. The right answer is 85.
Common mistakes: Circling Ones because 7 + 8 looks hard.; Writing 75 again.
Time for a middle child: about 5 min.

G3 **Write a sum with one mistake for a grown-up to find. Many answers are right.**

(a) My sum: ☐☐ + ☐☐ = ☐☐

(b) The right answer is ☐☐.

(c) The mistake is _____ .

Answer: Many answers are right: the sum has exactly one mistake and the right answer is given

Hint (say this first): Pick two numbers. Work out the right answer first. Then change one thing.

Worked solution: For example 36 + 27 = 53: the right answer is 63, and the mistake is that the ten was forgotten. Check: add back 53 − 27 = 26, not 36.

Common mistakes: Writing a sum with no mistake.; Changing two things at once.

Time for a middle child: about 5 min.

Investigation (page 5): The same mistake three times

Easy first step: A child writes 25 + 19 = 34. Check it: 34 − 19 = ? Is it right?

Then: The same child writes 36 + 27 = 53 and 48 + 35 = 73. What does the child forget each time? Write one more sum this child would get wrong.

Model answer: First: 34 − 19 = 15, not 25, so it is wrong. The right answer is 44. Each wrong answer is 10 less than the right one (44, 63 and 83). The child forgets the ten made by adding the ones. Another sum the child gets wrong needs 10 or more in the ones, for example 29 + 14 = 33 (right: 43).

Look for: A child who finds that every wrong answer is exactly 10 less than the right one, or who says the child forgets the carried ten. Credit any clear reason. A new sum must make 10 or more in the ones; a sum with no carry is not a good example.

Common mistakes: Saying the child cannot add.; Writing a new sum with no ten to carry.

Talk and explain (minutes 70 to 76)

A child in another class writes 45 + 28 = 63. She works it out again the same way and gets 63 again. She says it must be right. Do you agree? How could she check? Sentence starter: I think ... because ...

Model answer: No. Doing the sum the same way again can make the same mistake again. A different way, such as taking 28 away from 63, gives 35 and not 45, so 63 is wrong. The right answer is 73.

Look for: A child who says to check a different way, or who adds back. Not just yes or no. Volunteers first, a pass is always fine, never call a child by name to answer, no tallies.

Exit ticket (minutes 76 to 82): 3 questions

E1 **A child writes 63 − 28 = 45.**

(a) Check: 45 + 28 = ☐☐

(b) Is the answer right? Circle Yes or No.

Answer: 73; No

Worked solution: 45 + 28 = 73, not 63. So 45 is wrong. The right answer is 35, because 35 + 28 = 63.

Common mistakes: Saying Yes because 45 looks right.; Writing 63 for part a.

E2 **A child works out 28 + 15 in steps. Which step is wrong? Write its number in the bracket.**

Answer: (2) 15 − 2 = 12

Worked solution: 15 split into 2 and 13. 15 − 2 = 13, not 12. Then 30 + 13 = 43.

Common mistakes: Choosing (3) because it is the last step.; Choosing (1) because 28 + 2 looks odd.

E3 **Lily has $20. She spends $12. A child says she has $9 left.**

(a) Is the child right? Circle Yes or No.

(b) Lily has $____ left.

Answer: No; $8

Worked solution: Count up from 12 to 20: 8. Check: 8 and 12 make 20. The child's 9 and 12 make 21, so 9 is wrong.

Common mistakes: Saying Yes.; Adding 20 and 12.

Look-back (minutes 82 to 85)

Children tick or write: *Something I did well*; *The hardest part was* (checking a different way; finding the wrong step; the time or money story; something else); *What helped me* (the cards; adding back; checking my answer; I was not stuck); *Next time I will try* (adding back; checking a different way; explaining why; asking a question); *I am still stuck (my grown-up will photograph this page for my tutor)* and *I would like to pass*. Never collected or read out in front of the group. A child who ticks still stuck gets a check-in in the open session next lesson.

5. Home practice and online practice

#	Tier	Question	Answer
H1	Start here	A child writes 34 + 18 = 42. Is the answer right? Circle Yes or No. I know because ___.	No. 34 + 18 = 52.
H2	Start here	The bus leaves at 3:30. The trip takes half an hour. A child says it arrives at 3:00. Is the child right? Circle Yes or No. The bus arrives at ____ .	No. 4:00.
H3	Start here	Siti has $15. She buys a book for $6. A child says she has $11 left. Is the child right? Circle Yes or No. Siti has $____ left.	No. $9.
H4	Think harder	Two children work out 46 + 38. One writes 74. The other writes 84. Use taking away to check. Who is right? Circle 74 or 84.	84
H5	Start here	Line A is 14 cm long. Line B is 9 cm long. Line A is ____ cm longer than Line B.	5

Online practice (`out/l29/practice`, key `t3l29`): 11 items; 8 have a template that makes unlimited similar questions with a computed answer. Every template variant is checked by the validator against a brute-force oracle and the runner's own JavaScript. No timers, no scores, no streaks.

6. Parent line for the term report

Your child is learning to check an answer a second way and to find the one step that went wrong. This week: right and wrong sums, a wrong time and a wrong total.

Term line (plan, section 5): Your child is learning to give a reason, and to read time and money in real life. Add one named strength and one next step from your own observation. No marks, no ranking, no claim of exam results.

7. Slide list

#	Time	Slide	On screen	Speaker note
1	0–2	**Spot the Mistake** s-title	TODAY WE WILL: / Check an answer a different way. / Find the one step that went wrong.	Lesson 9 of Term 3, 90 minutes. Objective: check a take away by adding back, find the one wrong step in a worked sum, and spot a wrong time or total in a story. Plain statement for families: checking by adding back, finding a wrong step and wrong times and totals in stories are Primary 1 work with help; telling that an answer cannot be right by thinking about how big it must be (estimating), spotting the wrong digit in a column sum and writing a wrong sum for a partner go past the school Primary 1 syllabus on purpose. Time plan: opening 0 to 2, warm-up 2 to 8, sneak-peek 8 to 12, explore 12 to 23, mini-lesson 23 to 34, brain break 34 to 38, workbook 38 to 70, talk 70 to 76, exit ticket 76 to 82, look-back 82 to 85, flex time and goodbye 85 to 90 (the buffer: do not start anything new). Think harder, Go further and the Investigation are for every child to choose. Say the recording notice in one line: you and the screen only; a child may keep the camera off.
2	2–8	**Warm-up: add back** s-rem1	[picture: number_line] / 40 − 15 = 25. Check: 25 + 15 = ?	Adding back (Term 1 Lesson 7, taught again). On the line, 25 is the start; jumps of 10 and 5 reach the empty box. Say 40 take away 15 is 25, and to check we add the 15 back: 25 and 15 make 40. If we get back to the number we started with, the take away is right. Ask: what is 25 plus 15 (40). A child who is unsure counts on from 25 in tens and ones with fingers. Children answer aloud. No writing.
3	2–8	**Jump a ten** s-rem2	[picture: number_line] / 46 + 30. Jump three tens. Where do you land?	A tens jump (Term 1 Lesson 6, taught again): from 46, three jumps of ten reach 56, 66, 76. The ones stay the same. Ask: where do you land (76). Say: tens change, ones stay. A child who is unsure counts three jumps of ten on fingers. Children answer aloud. No writing.
4	2–8	**One hour later** s-rem3	[picture: clock] / [picture: clock] / It is 3:30. What time is it one hour later?	One hour later (Term 3 Lesson 3, taught again). The first clock shows 3:30. One hour later the short hand moves to the next number and the long hand stays on the 6: 4:30. Ask: what time is it one hour later (4:30). A child who is unsure counts the hours on the clock face, 3 to 4, with the long hand staying still. Children answer aloud. The second clock has no hands so the answer is not shown. No writing.
5	8–12	**Sneak-peek puzzles** s-sneak	Ideas we have not learnt yet. A guess is a great start. Nobody is marked. / PUZZLE 1: anyone can try: / A child in another class writes / 40 + 20 = 70. / Is the answer right? / Circle Yes or No. / PUZZLE 2: think: / A child in another class writes / 27 + 16 = 33. / Is the answer right? / I know because ___.	Puzzle 1 (40 + 20 = 70 is wrong, it is 60) is an access question. Puzzle 2 is a thinking question on the lesson idea: 27 + 16 = 33 is wrong (43) because the ten from 7 + 6 = 13 was forgotten. A child who says 'it is not right because 7 and 6 make 13' or who checks by taking away has done well. Say: two sneak-peek puzzles, ideas we have not learnt yet; a guess is a great start and I do not know is welcome; nobody is marked. Children write on their own paper. Nobody holds paper to the camera. There is no private chat. Answers: No; No (27 + 16 = 43). Do not reveal puzzle 2 until Talk time. Do not write scores or keep a list.
6	12–23	**Explore: right or wrong?** s-explore1	45 + 28 = 63 | 70 − 30 = 40 / 84 − 6 = 82 | 18 + 4 = 22 / 53 − 20 = 73 | 62 − 8 = 54 / 17 + 5 = 21 | 38 + 24 = 62 / Put each card on RIGHT or WRONG. Check it a different way.	Round 1, about 5 minutes. Each child has the 8 mistake cards and the RIGHT and WRONG mat. The slide shows the same 8 sums, without saying which are right. Children check each card a different way (adding back for a take away, taking away for an add, or counting on) and place it on RIGHT or WRONG. Four are right: 70 take away 30, 18 + 4, 62 take away 8 and 38 + 24. Four are wrong, each in exactly one way: 45 + 28 = 63 (the answer is 73), 84 take away 6 = 82 (78), 53 take away 20 = 73 (33) and 17 + 5 = 21 (22). The sorted answer is on the next slide. Ask: how did you check the first one? A child with no cards writes the 8 sums on paper and makes two lists. Nobody holds a card up to the camera: children say the sum and show a thumb for right or a flat hand for wrong.

7	12–23	**Four right, four wrong** s-explore1-ans	WRONG | RIGHT / 45 + 28 = 63 | 70 − 30 = 40 / 84 − 6 = 82 | 18 + 4 = 22 / 53 − 20 = 73 | 62 − 8 = 54 / 17 + 5 = 21 | 38 + 24 = 62 / How did you check? Adding back, taking away, counting on.	The sorted answer for round 1: WRONG are 45 + 28 = 63, 84 take away 6 = 82, 53 take away 20 = 73 and 17 + 5 = 21. RIGHT are 70 take away 30 = 40, 18 + 4 = 22, 62 take away 8 = 54 and 38 + 24 = 62. Say: a neat answer can still be wrong, so we check. Children tick their own sorting. Nobody holds cards up.
8	12–23	**Explore: fix the wrong cards** s-explore2	Take the 4 WRONG cards. / What went wrong in each one? / Write the right answer on your paper.	Round 2, about 4 minutes. Children take the 4 WRONG cards and say what went wrong and what the right answer is, then write the four right answers on paper. Right answers: 45 + 28 = 73 (5 and 8 make 13, and the ten was forgotten, so the answer is 10 too small); 84 take away 6 = 78 (4 is less than 6, so the ones were swapped round: 6 take away 4 is 2); 53 take away 20 = 33 (the answer 73 is more than 53, so a plus sign was used); 17 + 5 = 22 (one short, a counting slip). Accept any clear words. The table is on the next slide. Children say the right answer aloud together on a count of three.
9	12–23	**The right answers** s-explore2-ans	Wrong card | Right answer | What went wrong? / 45 + 28 = 63 | 73 | the ten was lost / 84 − 6 = 82 | 78 | the ones were swapped / 53 − 20 = 73 | 33 | a plus sign was used / 17 + 5 = 21 | 22 | one short	The right answers for round 2 and what went wrong in each: 73 (the ten was lost), 78 (the ones were swapped round), 33 (a plus sign was used instead of take away) and 22 (one short, a counting slip). Children tick their own page. Say: each wrong answer leaves a clue about what went wrong.
10	12–23	**What did we notice?** s-explore3	How far was each wrong answer from the right one? / Does the size of the mistake give a clue? / Is checking the same way again a good check?	Take one or two answers from volunteers. Key ideas: a wrong answer leaves a clue. 63 is 10 away from 73 (a lost ten). 82 is 4 away from 78 (swapped ones). 73 is 40 away from 33 (the wrong sign). 21 is 1 away from 22 (a counting slip). A different way of checking finds the mistake; the same way again can make the same mistake. Repeat a child answer in your own words so the recording picks it up. Do not give the rule before the children say it.
11	23–34	**Add back to check** s-m1	A child in another class writes 71 − 26 = 55. / Check: 55 + 26 = ?	Worked example. The child wrote 71 take away 26 equals 55. To check, add the 26 back to the answer: 55 and 26 make 81. We should get back to 71, but we get 81. So the answer is wrong. The right answer is 45, because 45 and 26 make 71. Ask: what is 55 plus 26 (81); do we get 71 (no); is the answer right (no). The answer is on the next slide. Say: if adding back does not give the number we started with, there is a mistake. Then the your-turn question: 60 take away 18 equals 52, check it. Answer: 52 + 18 = 70, not 60; the right answer is 42.
12	23–34	**Add back: 81, not 71** s-m1-ans	55 + 26 = 81. That is not 71. So 55 is wrong. 45 + 26 = 71, so the right answer is 45.	The answer for the example on the slide before. 55 + 26 = 81. That is not 71. So 55 is wrong. 45 + 26 = 71, so the right answer is 45. Read it aloud so that children can tick their own work, and say what the check showed.
13	23–34	**Check with a different way** s-m2	A child in another class writes 29 + 35 = 54. / [picture: number_line] / Jump on the line. Do you land on 54?	A different way to check an add. The child wrote 29 plus 35 equals 54. On the number line jump 30 to land on 59, then 5 more to land on 64. The line says 64, but the child wrote 54, so the two ways disagree and there is a mistake. Check the new answer by taking away: 64 take away 35 is 29. Ask: where do you land (64); does it match 54 (no). Say: two different ways must agree. If they do not, look for the mistake. Children answer aloud. The answer is on the next slide. A child who gets 54 again may have repeated the same slip: ask them to count the ones, 9 and 5 make 14.
14	23–34	**The two ways disagree** s-m2-ans	29 + 30 = 59. 59 + 5 = 64. The jump lands on 64, not 54. The right answer is 64.	The answer for the example on the slide before. 29 + 30 = 59. 59 + 5 = 64. The jump lands on 64, not 54. The right answer is 64. Read it aloud so that children can tick their own work, and say what the check showed.

15	23–34	**Which step is wrong?** s-m3	Step | Working / 1 | 56 + 20 = 76 / 2 | 76 + 4 = 80 / 3 | 80 + 3 = 82 / A child adds 56 + 27 in three steps. Which step is wrong?	Worked example. A child works out 56 plus 27 in three steps: 56 plus 20 is 76 (right); 76 plus 4 is 80 (right); 80 plus 3 is 82 (wrong, it is 83). Only step 3 is wrong. Ask: check each step in turn; which one is wrong (3). What is the right step (80 + 3 = 83). Then check the answer: 83 take away 27 is 56. Say: we find the one wrong step, we fix it, and the steps before it were fine. The answer is on the next slide. Your-turn question, children write: 92 take away 35 in two steps: step 1 92 take away 30 is 72; step 2 72 take away 5 is 67. Which step is wrong? Step 1 (92 take away 30 is 62), and step 2 is right working from the wrong number: fix step 1 and then do step 2 again: 62 take away 5 is 57. The answer is 57, check 57 + 35 = 92.
16	23–34	**Step 3 is wrong** s-m3-ans	Step 3 is wrong: 80 + 3 = 83. The right answer is 83. Check: 83 – 27 = 56.	The answer for the example on the slide before. Step 3 is wrong: 80 + 3 = 83. The right answer is 83. Check: 83 – 27 = 56. Read it aloud so that children can tick their own work, and say what the check showed.
17	23–34	**A wrong time** s-m4	[picture: clock] / [picture: clock] / The bus leaves at 9:00. The trip takes half an hour. A child says it arrives at 9:50. Is that right?	A wrong time. The bus leaves at 9 o'clock and the trip takes half an hour. A child says it arrives at 9:50. Ask: is that right (no). How can we check? Move the long hand from the 12 to the 6: that is half way round, 30 minutes: 9:30. Say: half an hour is 30 minutes; checking on a clock shows the mistake. The answer is on the next slide. A child who says 9:50 may think half of 100 is 50: ask them to count minutes in fives, 5, 10, 15, 20, 25, 30. Children answer aloud. The second clock has no hands so the answer is not shown.
18	23–34	**Half an hour is 30 minutes** s-m4-ans	Half an hour after 9:00 is 9:30. A half hour is 30 minutes, not 50. The bus arrives at 9:30.	The answer for the example on the slide before. Half an hour after 9:00 is 9:30. A half hour is 30 minutes, not 50. The bus arrives at 9:30. Read it aloud so that children can tick their own work, and say what the check showed.
19	23–34	**A wrong total** s-m5	[picture: price_tag] / [picture: price_tag] / A pen costs $4 and a book costs $9. A child says they cost $15 altogether. Is that right?	A wrong total. A pen costs 4 dollars and a book costs 9 dollars. A child says they cost 15 dollars altogether. Ask: is that right (no). Add 4 and 9: 9 and 1 make 10, and 3 more make 13. Check the child's total by taking the book away: 15 take away 9 is 6, but the pen costs 4, so 15 is not right. Say: we add prices to find the total, and we can check the total by taking one price away. The answer is on the next slide. Children answer aloud. A child who gives 15 may have added 4 and 9 and a slip of 2: ask them to count on from 9.
20	23–34	**13 dollars, not 15** s-m5-ans	4 + 9 = 13. The cost is $13, not $15. Check: 15 – 9 = 6, and 6 is not 4.	The answer for the example on the slide before. 4 + 9 = 13. The cost is $13, not $15. Check: 15 – 9 = 6, and 6 is not 4. Read it aloud so that children can tick their own work, and say what the check showed.
21	34–38	**Tick-tock break** s-break	Stand up. Swing your arms like the hands of a clock. / I say a sum. Say the right answer in a big voice. / Sip some water.	Four minutes. Stand up on camera with a little clear space. Swing both arms slowly like the hands of a clock: one hour is a big swing, half an hour is a half swing. Then call quick facts for children to answer aloud together: 10 plus 5, 20 plus 30, 50 take away 20, 7 and 3, 40 take away 10. Ask: is 6 plus 6 equals 13 right? Wrong! Say the right answer: 12. No equipment, no running. A child may stay seated and swing one arm. Do not give prizes or compare children.
22	38–70	**Workbook page 1: Start here** s-ws1	Questions 1 to 3. Take your time. / questions S1, S2, S3	Read questions 1 to 3 aloud. Children work on their own; the tutor walks the cameras and asks how do you know. Wait 10 seconds before helping. After question 3 any child may choose to carry on or try Think harder. Question 1: a take away, check by adding back, is it right. Question 2: the same with a different sum. Question 3: a worked sum in three steps, which step is wrong, and the right answer.
23	38–70	**Check page 1** s-ws1ans	answers S1, S2, S3	Read each answer aloud. Children tick their own page with a tick. Never ask a child to show a wrong answer to the group. Answers: Q1: 41, Yes. Q2: 60, No, 27. Q3: step 2, and 64.

24	38–70	**Workbook page 2: Start here** s-ws2	Questions 4 to 6. Take your time. / questions S4, S5, S6	Read questions 4 to 6 aloud. Question 4: a clock at 2:30, reads for one hour, a child says 3:00; is the child right and what time does he stop. Question 5: a bag and a cup, a child says $12 altogether. Question 6: tick True or False for four sums. Wait 10 seconds before helping.
25	38–70	**Check page 2** s-ws2ans	answers S4, S5, S6	Read each answer aloud so a child watching the recording can tick their own work. Never ask a child to show a wrong answer to the group. Answers: Q4 No; 3:30. Q5 No; $13. Q6 True, False, True, False.
26	38–70	**Choose your next page** s-choice	Page | What it is / Think harder (page 3) | Which step is wrong, two ways to check, fewer than, a clock / Go further (page 4) | Does it make sense? A column sum, your own puzzle / Investigation (page 5) | The same mistake three times / Check | Finish and check your own work / Everyone may choose. Not finished? Pages 3 to 5 are for home too.	Invite the whole group at once, for example: when you have finished Start here, you can choose Think harder, Go further or the Investigation. Never announce who is on which page and never name a child. Pairs are not split unless both children agree. Model answers are in the tutor guide. About 5 minutes.
27	70–76	**Talk and explain** s-talk	A CHILD IN ANOTHER CLASS SAYS: / 45 + 28 = 63. / I worked it out again the same way. / I got 63 again. So it must be right. / Do you agree? How could she check? / I think … because …	One minute of talk to the grown-up next to you, or to your screen. Volunteers first; a pass is always fine; never call a child by name to answer. Model answer: no, we do not agree. Doing it the same way again can make the same mistake again. A different way, such as taking away 28 from 63, gives 35, not 45, so 63 is wrong. The right answer is 73. Link to Explore: that was one of the wrong cards. Repeat each child answer in your own words for the recording. The child in the story is in another class: no cast name is given the wrong answer.
28	70–76	**Back to the sneak-peek** s-sneakans	Puzzle 1: No. 40 + 20 = 60. Puzzle 2: No. 27 + 16 = 43, because 7 + 6 = 13 and the ten was forgotten.	Reveal both puzzles now. Say: many of you wrote a good reason. Do not read out anyone earlier answer. Puzzle 1: 40 plus 20 is 60, so 70 is wrong. Puzzle 2: 27 plus 16 is 43, so 33 is wrong, 10 too small. A check that works: 33 take away 16 is 17, not 27. It was the same idea as Explore: check a different way, and look at how far the wrong answer is from the right one.
29	76–82	**Exit ticket** s-exit	On your own. No looking back. / questions E1, E2, E3	Children answer on page 5. Read E1, E2 and E3 aloud, and the three options of E2 (the figures on the slide are small). Children tick their own ticket against the next slide. Nobody holds paper to the camera and there is no private chat.
30	76–82	**Check the exit ticket** s-exitans	answers E1, E2, E3	Read each answer aloud. Children tick their own ticket with a tick. Answers: E1 73 then No; E2 option 2; E3 No then 8.
31	82–85	**Look-back** s-look	[look-back box]	Three minutes. Children tick or write the look-back box on page 5. A pass is always allowed. It is never marked, never collected in front of the group and never read out. If a child ticks I am still stuck, a grown-up may photograph that page after class and send it to the business WhatsApp (the number is in the parent summary, never on a slide). You then help in the open session next lesson. Never a direct message to a child. The Next time I will try row is optional.

32	85–90	**Well done, inquirers!** s-end	Show a grown-up one mistake you spotted today. / NEXT LESSON: / Calendar Detective, and Mi's Challenge Quest / Have ready: the Quest Log, closed. A quest partner. The 7 day cards, cut out by a grown-up.	Say goodbye and thank the grown-ups. Minutes 85 to 90 are flex time: the buffer. If you ran late, finish any open item (the look-back). Do not start anything new. Remind families that the next lesson is the last of Term 3 and holds Mi's Challenge Quest, a game with Mi: they need the Quest Log closed in its envelope, a quest partner (a grown-up or a brother or sister), and the calendar page with the 7 day cards cut out by a grown-up. If the whole class is ahead, show the optional slides and use the Spare-time menu from the tutor guide.
33	optional	**Go further: be the detective** s-opt	A child writes 8 + 7 = 14. She checks: 14 – 7 = 7. / Is the answer right? Does the check work?	Optional, only if you are on time. The sum 8 + 7 = 14 is wrong (15). The check 14 take away 7 is 7, not 8: so the check shows the mistake. Ask: does the check agree with the sum (no, 14 take away 7 is 7 but the first number was 8). So there is a mistake and the right answer is 15. Take two answers and move on. This links to adding back in the Mini-lesson.
34	optional	**If we have time: pick one** s-menu	Activity | What we do / Is it always true? | Does adding back always find a mistake in a take away? Test it. / Make a puzzle | Write a sum with one mistake for your grown-up. / Teach it back | Show a toy how to find the wrong step. / Optional. Everyone may join.	Optional Spare-time menu, used only if the whole class is ahead. Pick one or two activities (3 to 5 minutes each); full tutor lines and answers are in the tutor guide. Never start new content and never name who is ahead. Slow down with how do you know rather than rushing on. These minutes come from the flex buffer and are not part of the 90-minute sum.

8. Recording cues and chapter markers

Say each cue once, to the camera, at the start of the part. Add a chapter marker at each time. A child watching alone cannot ask you a question, so say the page and question number aloud, read each question and each answer aloud, and repeat a child's answer in your own words.

Chapter	Part	You say (recording voice)	A child watching later
0:00	Opening	*If you are watching the recording: hello! Get a pencil, paper, your workbook and your mistake cards. Pause or rewind any time.*	Gather your workbook, mistake cards, pencil and paper. Press play when ready.
2:00	Remember warm-up	*Recording: join in! Look at each picture and say your answer out loud. Pause if you need longer.*	Play along. Say each answer out loud before the video does.
8:00	Sneak-peek puzzles	*Recording: pause now and try both puzzles on your paper. Guessing is fine. Press play for the next part.*	Pause. Try both puzzles on your own paper. A guess is fine.
12:00	Explore	*Recording: pause now. Sort your 8 cards into RIGHT and WRONG. Then fix each wrong card. Press play when you have tried.*	Pause. Check each card a different way, sort the 8 cards, and fix the wrong ones.
23:00	Mini-lesson	*Recording: keep watching and copy the examples. Rewind if you miss a step.*	Watch and copy the examples. Rewind any step you missed.
34:00	Brain break	*Recording: stand up and join in! Tick-tock with your arms, then say the right answer with a big voice.*	Stand up. Swing your arms like a clock hand and call the answers, then sit down.
38:00	Workbook	*Recording: pause now and do workbook page 1. When you finish a page, press play to check it.*	Pause. Do page 1, then play to check. Then page 2. Then choose a page to try.
70:00	Talk time	*Recording: pause. Tell a grown-up whether checking the same way again is a good check, and why.*	Pause. Explain your answer to someone at home and show how you know.

76:00	Exit ticket	*Recording: pause and do the exit ticket on your own. Then press play to check it.*	Pause. Do the exit ticket alone, then play to check it.
82:00	Look-back	*Recording: pause and fill in your look-back box. You can pass.*	Pause. Fill in the look-back box, or pass.
85:00	Flex time and goodbye	*Recording: well done for finishing! Show a grown-up one mistake you spotted today.*	Show a grown-up your workbook and tell them one mistake you spotted.

9. Timing check (model, not a stopwatch)

Part	Allocated	Script	Slow class	Flag
opening	2:00	1:45	2:00	ok
remember	6:00	4:49	5:43	ok
sneak	4:00	3:10	3:43	ok
explore	11:00	8:55	10:35	ok
mini	11:00	8:52	10:22	ok
break	4:00	3:13	3:49	ok
work	32:00	24:42	29:21	ok
talk	6:00	4:30	5:20	ok
exit	6:00	4:49	5:38	ok
look	3:00	2:06	2:27	ok
flex	5:00	0:34	0:39	ok
Total	90:00	67:25	79:37	

Assumptions as docs/audit/timing_p1.md: speech 120 words a minute, read-aloud 110, slide change 25 s, a group question 10 s thinking plus 20 s answers, papers to the camera 30 s. Workbook: Start here has 6 questions; a middle child needs about 9.5 min for page 1 and 9.0 min for page 2 (a slower child about 1.7 times as long, so may finish page 1 and part of page 2; that is fine, the check-together slides cover every answer). The lesson has no breakout rooms, so no breakout time is counted.